

Администрирование сетевых подсистем

Лабораторная работа №11

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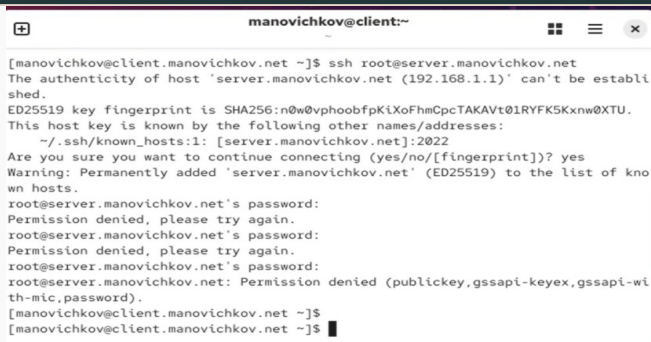
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Цели и задачи работы

Приобретение практических навыков по настройке безопасного удалённого доступа к серверу с помощью протокола SSH.

Выполнение лабораторной работы

Запрет удалённого доступа root



```
manovichkov@client:~  
[manovichkov@client.manovichkov.net ~]$ ssh root@server.manovichkov.net  
The authenticity of host 'server.manovichkov.net (192.168.1.1)' can't be established.  
ED25519 key fingerprint is SHA256:n0w0vp0oobfpKiXoFhmCpcTAKAVt01RYFK5Kxw0XTU.  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:1: [server.manovichkov.net]:2022  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'server.manovichkov.net' (ED25519) to the list of known hosts.  
root@server.manovichkov.net's password:  
Permission denied, please try again.  
root@server.manovichkov.net's password:  
Permission denied, please try again.  
root@server.manovichkov.net's password:  
root@server.manovichkov.net: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).  
[manovichkov@client.manovichkov.net ~]$  
[manovichkov@client.manovichkov.net ~]$
```

Рис. 1: Попытка входа под root

Проверка параметра PermitRootLogin

```
root@server:~ - sudo -i
sshd_config  [-M--] 18 L:[ 20+20  40/131] *(1154/3658b) 0010 0x00A
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

Повторная попытка входа

```
[manovichkov@client.manovichkov.net ~]$ ssh root@server.manovichkov.net
root@server.manovichkov.net's password:
Permission denied, please try again.
root@server.manovichkov.net's password:
Permission denied, please try again.
root@server.manovichkov.net's password:
root@server.manovichkov.net: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).
[manovichkov@client.manovichkov.net ~]$
```

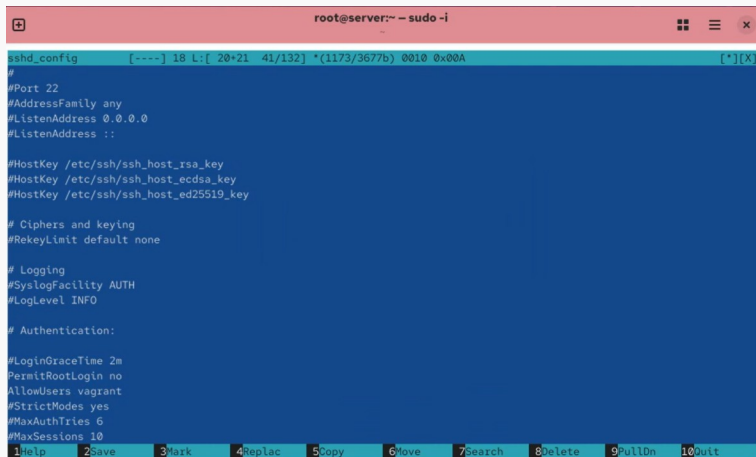
Рис. 3: Отклонение доступа root

```
[manovichkov@client.manovichkov.net ~]$ ssh manovichkov@server.manovichkov.net
manovichkov@server.manovichkov.net's password:
Web console: https://server.manovichkov.net:9090/ or https://192.168.1.1:9090/

Last login: Tue Oct 21 10:12:17 2025
[manovichkov@server.manovichkov.net ~]$
logout
Connection to server.manovichkov.net closed.
```

Рис. 4: Подключение manovichkov

Ограничение списка пользователей SSH



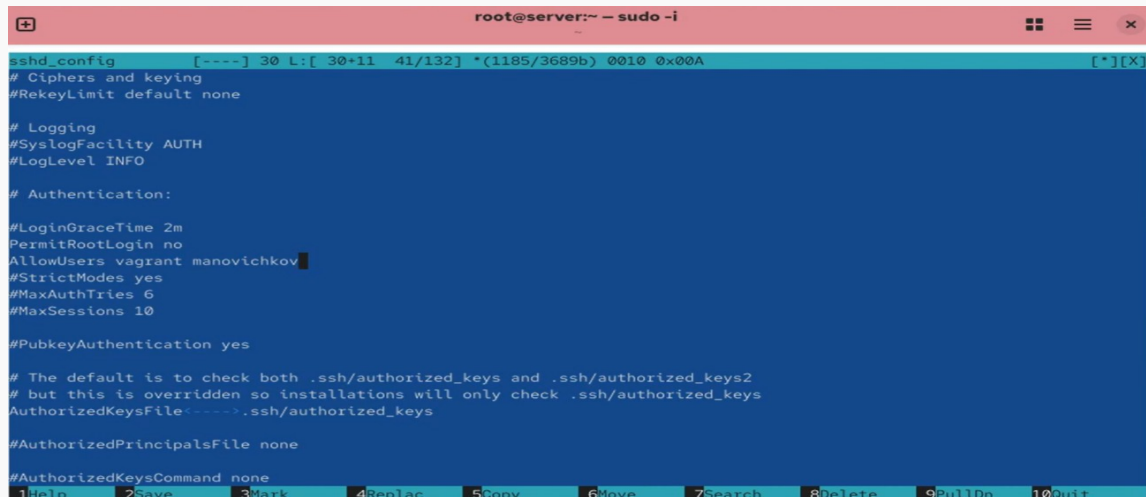
The image shows a terminal window with a red title bar that reads "root@server:~ -- sudo -i". The terminal content displays the configuration for the sshd_config file. The configuration includes settings for port (22), listen address (0.0.0.0), host keys, ciphers, logging, and authentication. The "Authentication:" section is highlighted, showing the following parameters:

```
# Authentication:
#LoginGraceTime 2m
PermitRootLogin no
AllowUsers vagrant
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

At the bottom of the terminal, a navigation bar is visible with the following options: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, 10Quit.

Рис. 5: Параметр AllowUsers vagrant

Расширение списка разрешённых пользователей



A terminal window titled "root@server:~ - sudo -i" displays the contents of the `sshd_config` file. The window has a red title bar and standard window controls. The terminal output shows various configuration options, with the `AllowUsers` line set to `vagrant manovichkov`. At the bottom, a status bar lists keyboard shortcuts: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, 10Quit.

```
root@server:~ - sudo -i
sshd_config      [----] 30 L:[ 30+11 41/132] *(1185/3689b) 0010 0x00A [*][X]
# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
AllowUsers vagrant manovichkov
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile<---->.ssh/authorized_keys

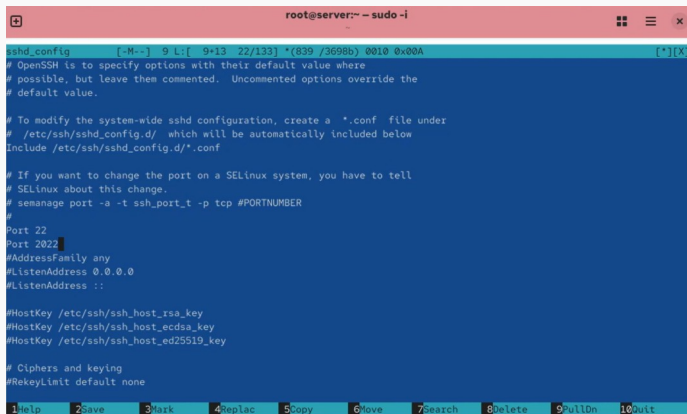
#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
1Help      2Save      3Mark      4Replac    5Copy      6Move      7Search    8Delete    9PullDn    10Quit
```

```
[manovichkov@client.manovichkov.net ~]$ ssh manovichkov@server.manovichkov.net
manovichkov@server.manovichkov.net's password:
Web console: https://server.manovichkov.net:9090/ or https://192.168.1.1:9090/

Last failed login: Tue Oct 21 10:21:43 UTC 2025 from 192.168.1.30 on ssh:notty
There were 3 failed login attempts since the last successful login.
Last login: Tue Oct 21 10:20:34 2025 from 192.168.1.30
[manovichkov@server.manovichkov.net ~]$
logout
Connection to server.manovichkov.net closed.
```

Добавление дополнительных портов SSH



```
root@server:~ -- sudo -i
sshd_config  [-M--]  9 L:[ 9+13 22/133] *(839 /3698b) 0010 0x00A  [*][X]
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

# To modify the system-wide sshd configuration, create a *.conf file under
# /etc/ssh/sshd_config.d/ which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf

# If you want to change the port on a SELinux system, you have to tell
# SELinux about this change.
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
Port 22
Port 2022
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

1Help 2Save 3Mark 4Replac 5Copy 6love 7Search 8Delete 9ullDn 10Quit
```

Рис. 8: Добавление портов 22 и 2022

Ошибка привязки порта SFI linux

```
[root@server.manovichkov.net ~]# systemctl restart sshd
[root@server.manovichkov.net ~]# systemctl status -l sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-10-21 10:24:45 UTC; 21s ago
 Invocation: 8b8ae7304cb64131a648f9fe8984a6ea
    Docs: man:sshd(8)
          man:sshd_config(5)
 Main PID: 11755 (sshd)
   Tasks: 1 (limit: 10381)
  Memory: 1M (peak: 1.2M)
     CPU: 5ms
   CGroup: /system.slice/ssh.service
           └─11755 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Oct 21 10:24:45 server.manovichkov.net systemd[1]: Starting sshd.service - OpenSSH server daemon...
Oct 21 10:24:45 server.manovichkov.net (sshd)[11755]: sshd.service: Referenced but unset environment variable evaluat
Oct 21 10:24:45 server.manovichkov.net sshd[11755]: error: Bind to port 2022 on 0.0.0.0 failed: Permission denied.
Oct 21 10:24:45 server.manovichkov.net sshd[11755]: error: Bind to port 2022 on :: failed: Permission denied.
Oct 21 10:24:45 server.manovichkov.net sshd[11755]: Server listening on 0.0.0.0 port 22.
Oct 21 10:24:45 server.manovichkov.net sshd[11755]: Server listening on :: port 22.
Oct 21 10:24:45 server.manovichkov.net systemd[1]: Started sshd.service - OpenSSH server daemon.
[root@server.manovichkov.net ~]#
```

Рис. 9: Ошибка порта
2022

Исправление SELinux и firewall

```
[root@server.manovichkov.net ~]# firewall-cmd --add-port=2022/tcp
success
[root@server.manovichkov.net ~]# firewall-cmd --add-port=2022/tcp --permanent
success
[root@server.manovichkov.net ~]# systemctl restart sshd
[root@server.manovichkov.net ~]# systemctl status -l sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-10-21 10:26:44 UTC; 11s ago
 Invocation: 4ad7af504a7e4709b8ab77bc882afa2a
    Docs: man:sshd(8)
          man:sshd_config(5)
 Main PID: 12031 (sshd)
   Tasks: 1 (limit: 10381)
  Memory: 1M (peak: 1.2M)
    CPU: 4ms
  CGroup: /system.slice/sshd.service
          └─12031 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Oct 21 10:26:44 server.manovichkov.net systemd[1]: Starting sshd.service - OpenSSH server daemon...
Oct 21 10:26:44 server.manovichkov.net (sshd)[12031]: sshd.service: Referenced but unset environment variable evaluated
Oct 21 10:26:44 server.manovichkov.net sshd[12031]: Server listening on 0.0.0.0 port 2022.
Oct 21 10:26:44 server.manovichkov.net sshd[12031]: Server listening on :: port 2022.
Oct 21 10:26:44 server.manovichkov.net sshd[12031]: Server listening on 0.0.0.0 port 22.
Oct 21 10:26:44 server.manovichkov.net sshd[12031]: Server listening on :: port 22.
Oct 21 10:26:44 server.manovichkov.net systemd[1]: Started sshd.service - OpenSSH server daemon.
```

Рис. 10: Прослушивание портов 22 и 2022

Проверка подключений по двум портам

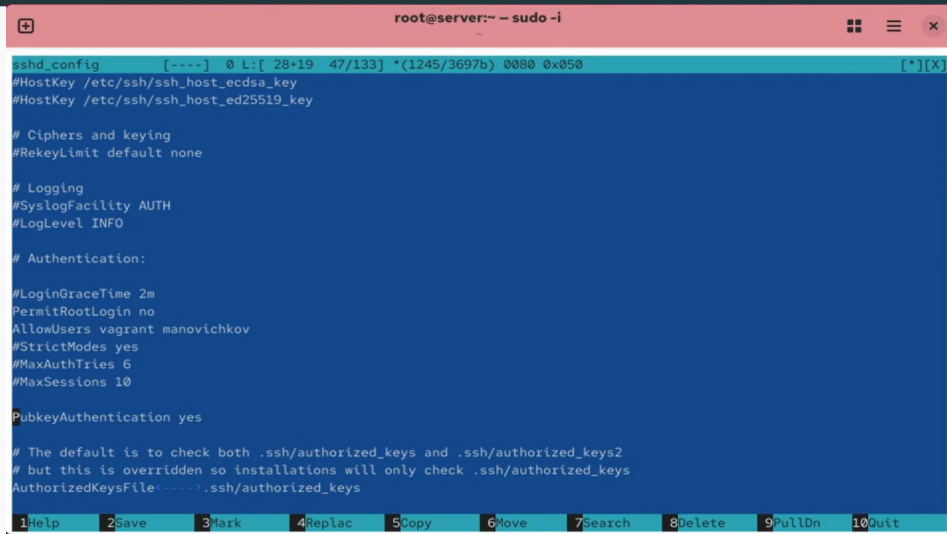
```
[manovichkov@client.manovichkov.net ~]$ ssh manovichkov@server.manovichkov.net
manovichkov@server.manovichkov.net's password:
Web console: https://server.manovichkov.net:9090/ or https://192.168.1.1:9090/

Last failed login: Tue Oct 21 10:21:43 UTC 2025 from 192.168.1.30 on ssh:notty
There were 3 failed login attempts since the last successful login.
Last login: Tue Oct 21 10:20:34 2025 from 192.168.1.30
[manovichkov@server.manovichkov.net ~]$
logout
Connection to server.manovichkov.net closed.
[manovichkov@client.manovichkov.net ~]$
[manovichkov@client.manovichkov.net ~]$
[manovichkov@client.manovichkov.net ~]$ ssh manovichkov@server.manovichkov.net
manovichkov@server.manovichkov.net's password:
Web console: https://server.manovichkov.net:9090/ or https://192.168.1.1:9090/

Last login: Tue Oct 21 10:22:41 2025 from 192.168.1.30
[manovichkov@server.manovichkov.net ~]$ sudo -i
[sudo] password for manovichkov:
[root@server.manovichkov.net ~]#
```

Рис. 11: Подключение по портам 22 и 2022

Разрешение аутентификации по ключу



A terminal window titled "root@server:~ -- sudo -i" displays the configuration file /etc/ssh/sshd_config. The window has a pink title bar and a blue background. The configuration text is as follows:

```
sshd_config      [----] 0 L:[ 28+19 47/133] *(1245/3697b) 0080 0x050 [*][X]
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
AllowUsers vagrant manovichkov
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile<---->.ssh/authorized_keys
```

At the bottom of the terminal, there is a status bar with the following shortcuts: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, 10Quit.

Проверка входа по ключу

```
manovichkov@client:~  
|      oooo      |  
+----[SHA256]-----+  
[manovichkov@client.manovichkov.net ~]$  
[manovichkov@client.manovichkov.net ~]$ ssh-copy-id manovichkov@server.manovichkov.net  
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L  
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed  
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys  
manovichkov@server.manovichkov.net's password:  
  
Number of key(s) added: 1  
  
Now try logging into the machine, with: "ssh 'manovichkov@server.manovichkov.net'"  
and check to make sure that only the key(s) you wanted were added.  
  
[manovichkov@client.manovichkov.net ~]$ ssh manovichkov@server.manovichkov.net  
Web console: https://server.manovichkov.net:9090/ or https://192.168.1.1:9090/  
  
Last login: Tue Oct 21 10:28:34 2025 from 192.168.1.30  
[manovichkov@server.manovichkov.net ~]$  
logout  
Connection to server.manovichkov.net closed.  
[manovichkov@client.manovichkov.net ~]$
```

Рис. 13: Подключение без пароля

```
[manovichkov@client.manovichkov.net ~]$ lsof | grep TCP
[manovichkov@client.manovichkov.net ~]$ ssh -fNL 8080:localhost:80 manovichkov@s
erver.manovichkov.net
[manovichkov@client.manovichkov.net ~]$ lsof | grep TCP
ssh          11636                  manovichkov      3u      IPv4          91488
      0t0      TCP client.manovichkov.net:60038->ns.manovichkov.net:ssh (ESTAB
LISHED)
ssh          11636                  manovichkov      4u      IPv6          91495
      0t0      TCP localhost:webcache (LISTEN)
ssh          11636                  manovichkov      5u      IPv4          91496
      0t0      TCP localhost:webcache (LISTEN)
[manovichkov@client.manovichkov.net ~]$
```

Рис. 14: Создание туннеля 8080 → 80

Проверка туннеля в браузере

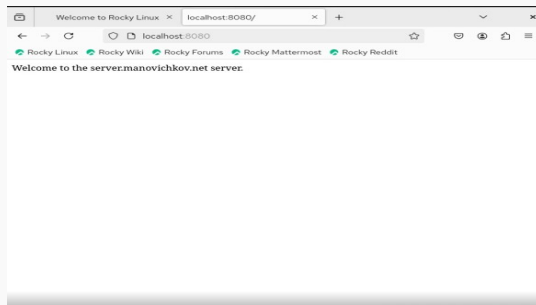


Рис. 15: Страница приветствия сервера

Запуск консольных приложений через SSH

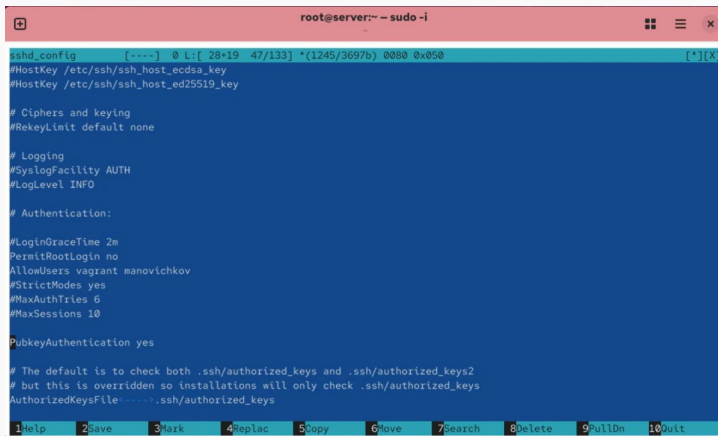
```
drwxr-xr-x. 2 manovichkov manovichkov 6 Sep 13 06:18 Downloads
drwx-----, 4 manovichkov manovichkov 32 Sep 13 06:18 .local
drwx-----, 5 manovichkov manovichkov 4096 Oct 17 10:03 Maildir
drwxr-xr-x. 5 manovichkov manovichkov 54 Sep 17 06:57 .mozilla
drwxr-xr-x. 2 manovichkov manovichkov 6 Sep 13 06:18 Music
drwxr-xr-x. 2 manovichkov manovichkov 6 Sep 13 06:18 Pictures
drwxr-xr-x. 2 manovichkov manovichkov 6 Sep 13 06:18 Public
drwx-----, 2 manovichkov manovichkov 29 Oct 21 10:31 .ssh
drwxr-xr-x. 2 manovichkov manovichkov 6 Sep 13 06:18 Templates
-rw-r-----, 1 manovichkov manovichkov 6 Oct 21 10:12 .vboxclient-clipboard-t
ty2-control.pid
-rw-r-----, 1 manovichkov manovichkov 6 Oct 21 10:36 .vboxclient-clipboard-t
ty2-service.pid
-rw-r-----, 1 manovichkov manovichkov 6 Oct 21 10:12 .vboxclient-draganddrop
-tty2-control.pid
-rw-r-----, 1 manovichkov manovichkov 6 Oct 21 10:12 .vboxclient-hostversion
-tty2-control.pid
-rw-r-----, 1 manovichkov manovichkov 6 Oct 21 10:12 .vboxclient-seamless-tt
y2-control.pid
-rw-r-----, 1 manovichkov manovichkov 6 Oct 21 10:12 .vboxclient-vmsvga-sess
ion-tty2-control.pid
-rw-r-----, 1 manovichkov manovichkov 5 Oct 21 10:12 .vboxclient-vmsvga-sess
ion-tty2-service.pid
drwxr-xr-x. 2 manovichkov manovichkov 6 Sep 13 06:18 Videos
```

Рис. 16: Вывод hostname и ls

```
[manovichkov@client.manovichkov.net ~]$ ssh manovichkov@server.manovichkov.net M
AIL=~/.Maildir mail
s-nail version v14.9.24.  Type '?' for help
/home/manovichkov/Maildir: 3 messages
▶  1 manovichkov          2025-10-13 07:12    18/685    "hello"
   2 manovichkov@client.m 2025-10-17 09:29    21/874    "LMTP test"
   3 manovichkov          2025-10-17 10:03    22/861    "hello"
q
```

Рис. 17: Просмотр почты Maildir

Включение X11Forwarding



A terminal window titled "root@server:~ - sudo -i" displays the contents of the `sshd_config` file. The window has a pink title bar and a blue background. The configuration file content is as follows:

```
sshd_config      [----] 0 L: [ 28+19 47/133] *(1245/3697b) 0080 0x050 [*][X]
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
AllowUsers vagrant manovichkov
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile----- .ssh/authorized_keys
```

At the bottom of the terminal, there is a navigation bar with the following shortcuts: 1 Help, 2 Save, 3 Mark, 4 Replac, 5 Copy, 6 Move, 7 Search, 8 Delete, 9 PullDn, 10 Quit.

Неудачная попытка запуска Firefox

```
[manovichkov@client.manovichkov.net ~]$ ssh -YC manovichkov@server.manovichkov.net firefox
Warning: No xauth data; using fake authentication data for X11 forwarding.
X11 forwarding request failed on channel 0
Error: no DISPLAY environment variable specified
[manovichkov@client.manovichkov.net ~]$ ssh -YC manovichkov@server.manovichkov.net firefox
Warning: No xauth data; using fake authentication data for X11 forwarding.
X11 forwarding request failed on channel 0
Error: no DISPLAY environment variable specified
[manovichkov@client.manovichkov.net ~]$ █
```

Рис. 19: Ошибка X11
отображения

Выводы по проделанной работе

В ходе лабораторной работы были освоены приёмы конфигурирования SSH-сервера. Настроены параметры безопасности, включая запрет root-доступа и ограничение списка пользователей.

Добавлены дополнительные порты, произведена настройка SELinux и межсетевого экрана. Реализована аутентификация по ключам, локальное перенаправление портов и создание SSH-туннелей.

Проверена работа консольных и графических приложений, а также подготовлена структура для автоматического применения конфигурации.